Ghislain THEUBO

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Professional Summary

Embedded-oriented Software Engineer (Ensimag) with hands-on **C/FreeRTOS** real-time control development at **INRIA** (electrically assisted wheelchair — sensors, latency, user validation) and strong systems discipline from industrial R&D. Proficient in C/C++, comfortable with Python, Git, and rigorous coding standards. Motivated by mission-critical autonomy systems. Seeking **Embedded Software Engineer** at Harmattan AI (Paris, on-site) to contribute to UAV embedded Linux/RTOS software, sensor integration, and telemetry.

Core Strengths

C/C++ · **FreeRTOS / real-time** · Embedded systems internship (INRIA) · Sensors & pipelines · Python plus · Debugging under constraints · Architecture & reliability mindset · English · Mission ownership

Experience

Shift Technology – Platform Software Engineering — Paris, France
Software Engineer (CDI – Platform / Backend) — *Sept 2026 – Nov 2026*

- Contributed to the **platform / foundations** squad building shared services used by product teams: design and implementation of **C# / .NET** components and **high-performance APIs** on **Microsoft Azure**.
- Worked with cloud data and messaging building blocks (**Azure Cosmos DB**, **Service Bus**), configuration/transversal services, and infrastructure-as-code awareness (**Terraform**) in a hybrid SaaS environment.
- Practiced production engineering habits: design discussions, code reviews, monitoring mindset, incremental delivery, and collaboration with platform and product engineers.

Gustave Roussy & CentraleSupélec MICS Lab — Paris / Saclay, France
Software Engineer (CDD – Backend / Data Platforms) — *Jan 2026 – Sept 2026*

- Software Engineer (CDD) in joint coordination between **Gustave Roussy (Oncostat / CESP)** and **CentraleSupélec MICS**: built and maintained production backends for research/clinical data workflows.
- Designed **REST APIs** (Python/FastAPI) with validation, RBAC, structured logging, and observability; PostgreSQL modeling with indexing and query tuning (**∼30%** faster critical reads).
- Containerized services with **Docker**; automated tests and delivery via **GitHub Actions**; wrote technical documentation for data flows and operational handoff.
- Collaborated daily with domain experts (research/medical) — translating ambiguous needs into reliable, auditable software.

Dassault Systèmes – 3DEXPERIENCE / 3DSwym R&D — Vélizy-Villacoublay, France
Software Engineer Intern (Full-Stack Module Rewrite) — *Mar 2025 – Sept 2025*

- Full-stack overhaul of the legacy **Survey module** for **3DSwym** (3DEXPERIENCE collaborative platform): analyze, design, develop, and deliver a modern replacement.
- Backend in **Java 17+ / Spring Boot** (MVC, Spring Data JPA) + **Hibernate**; applied **TDD** with JUnit; design patterns for modular maintainability.
- User-needs analysis with product "Champions", competitive benchmark, MoSCoW prioritization in Agile ceremonies; Git + CI/CD practices.
- Results: delivered core features of the new module with **∼35%** performance improvement and **∼85%** test coverage — robust, maintainable codebase for 3DSwym users.

INRIA Grenoble – Humanlab / Froll'N'Roll — Montbonnot-Saint-Martin, France
Research Intern – Real-Time Control Modules (Assistive Wheelchair) — *May 2024 – Aug 2024*

- Developed real-time control modules in **C/FreeRTOS** and Python for an electrically assisted manual wheelchair — direct user impact with continuous validation with the end user.
- Redesigned communication/data pipelines under latency constraints; improved responsiveness by **∼40%**; contributed to open/documented research codebases.
- Human-centered engineering: DIY hardware/software integration, iterative prototyping, and clear technical communication in a multidisciplinary SED team.

LIG – Laboratoire d'Informatique de Grenoble — Grenoble, France
Cloud Infrastructure Intern (VM Migration on OpenStack / Grid'5000) — *Feb 2025 – Mar 2025*

- Automated and studied **VM migration** experiments on OpenStack within Grid'5000; analyzed performance vs availability trade-offs for hosted applications.
- Wrote reproducible scripts and measurement notes under supervision of Alain Tchana — deepened virtualization and distributed cloud skills.

Selected Projects

INRIA Froll'N'Roll – Assisted Wheelchair Control | C, FreeRTOS, Python — INRIA

- Real-time control modules on embedded constraints; sensor data processing; +40% responsiveness via protocol/pipeline redesign; continuous validation with end user.
- Multidisciplinary collaboration (hardware/algorithms/users) — closest analog to UAV embedded + algorithmic team partnership.

Embedded Bring-up Notes / Driver Sketch | C/C++, datasheet-driven — GitHub

- Practice reading datasheets, sketching sensor driver interfaces, and documenting RTOS task design — ramp for Harmattan sensor integration.

C++ Systems / Mesh Library | C++, templates, Catch2 — GitHub

- Performance-minded C++ craftsmanship and testing habits transferable to embedded quality standards.

Technical Skills

Embedded: C, C++, FreeRTOS, multi-thread/process notions, Git, build systems
Plus: Python, Linux, profiling/debugging under latency constraints
Domain signal: INRIA assistive robotics/real-time; eagerness for Embedded Linux/RTOS UAV work
Languages: English professional; French fluent

Education & Certifications

Grenoble INP - Ensimag — Grenoble, France
Engineering Degree – Information Systems / Software Engineering (Double Degree) — *2023 – 2025*

École Nationale Supérieure Polytechnique de Yaoundé — Yaoundé, Cameroon
Master's Degree in Computer Engineering – Top 12% of class — *2019 – 2023*

Certifications: Programming Foundations – TDD (LinkedIn, 2025) · Primerli – Insurance Industry Foundations (2025)

Languages & Extra

English: bilingual / professional · **French:** fluent · **Spanish:** intermediate
Extra: academic tutoring (UGA DAEU maths; 4 years mentoring baccalaureate students in Cameroon) · tennis, football